

Introduction to Differential Equations

Differential Equation:

A differential equation is, in simpler terms, a statement of equality having a derivative or differentials.

An equation involving differentials or differential co-efficient is called a differential equation.

For examples, $\frac{d^2 y}{d x^2} = 0$ and $y dx + x dy = 0$ are two differential equations.

Ordinary Differential Equation:

If a differential equation contains one dependent variable and one independent variable, then the differential equation is called ordinary differential equation.

For example, (i) $\frac{dy}{d x} = x \sin x$, (ii) $4 \frac{d^2 y}{d x^2} + 6 y = \tan x$.

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Partial Differential Equation:

If there are two or more independent variables, so that the derivatives are partial, then the differential equation is called partial differential equation.

For example, $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = z$

Application of differential equations :

The differential equations for an LRC circuit, a pendulum, and a diffusing dye are given by

$$\begin{aligned} L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q &= \mathcal{E}_0 \cos \omega t, & (\text{RLC circuit equation}) \\ ml \frac{d^2 \theta}{dt^2} + cl \frac{d\theta}{dt} + mg \sin \theta &= F_0 \cos \omega t, & (\text{pendulum equation}) \\ \frac{\partial u}{\partial t} &= D \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right). & (\text{diffusion equation}) \end{aligned}$$

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Order: The order of an ordinary differential equations is the order of the highest order derivative

Examples :

$$\left(\frac{dy}{dx}\right) - y = e^x$$

First order ODE

$$\left(\frac{d^2y}{dx^2}\right) - 5\frac{dy}{dx} + 2y = \cos(x)$$

Second order ODE

$$\left(\frac{d^3y}{dx^3}\right)^2 - \frac{d^2y}{dx^2} + 2y^4 = 1$$

Third order ODE

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Degree: The **degree** of a differential equation is the highest degree of the highest derivative after the equation has been put in the form free from radicals and fraction.

Examples :

$$\left(\frac{dy}{dx}\right) - (y)^4 = e^x$$

1st order 1st degree ODE

$$\left(\frac{d^2y}{dx^2}\right)^{(3)} - 5\left(\frac{dy}{dx}\right)^4 + 2y = \cos(x)$$

2nd order 3rd degree ODE

$$\left(\frac{d^3y}{dx^3}\right)^{(2)} - \frac{dy}{dx} + 2y^4 = 1$$

3rd order 2nd degree ODE

Example 1

➤ What is the **degree** of a differential $\sqrt[4]{\frac{d y}{d x} + 2x\left(\frac{d^4 y}{d x^4}\right)^3} = \sqrt[3]{x-2} \quad ?$

Solution:

$$\sqrt[4]{\frac{d y}{d x} + 2x\left(\frac{d^4 y}{d x^4}\right)^3} = \sqrt[3]{x-2}$$

$$\text{or, } \left\{ \frac{d y}{d x} + 2x\left(\frac{d^4 y}{d x^4}\right)^3 \right\}^{\frac{1}{4}} = (x-2)^{\frac{1}{3}}$$

$$\text{or, } \left\{ \frac{d y}{d x} + 2x\left(\frac{d^4 y}{d x^4}\right)^3 \right\}^3 = (x-2)^4$$

$$\text{or, } \left(\frac{d y}{d x}\right)^3 + \dots\dots\dots + 8x^3\left(\frac{d^4 y}{d x^4}\right)^9 = (x-2)^4$$

4th order **9th** degree ODE

Classification of Differential Equ.

Linear: An ODE is linear if the unknown function and its derivatives appear to power one. No product of the unknown function and/or its derivatives.

Examples:

$$\frac{dy}{dx} - y = e^x$$

Linear ODE

$$\frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} + 2x^2 y = \cos(x)$$

Linear ODE

$$\left(\frac{d^2 y}{dx^2} \right)^3 - \frac{dy}{dx} + \sqrt{y} = 1$$

Non-linear ODE

Classification of Differential Equ.

Initial value and Boundary-Value Problems

Initial-Value Problems

The auxiliary conditions are at *one point of the independent variable*

$$y'' + 2y' + y = e^{-2x}$$

$$y(0) = 1, \quad y'(0) = 2.5$$

same

Boundary-Value Problems

The auxiliary conditions are *not at one point of the independent variable*

More difficult to solve than initial value problem

$$y'' + 2y' + y = e^{-2x}$$

$$y(0) = 1, \quad y(2) = 1.5$$

different

Solution of ODE:

General Solution of ODE:

The solution of a differential equation in which the number of arbitrary constants is equal to the order of the differential equation is called the general solution.

For example, $y = ax + b$ is the general solution of the differential equation $\frac{d^2y}{dx^2} = 0$

where a and b are arbitrary constants.

Particular Solution of ODE:

If particular values are given to the arbitrary constants in the general solution, then the solution so obtained is called particular solution.

For example, putting $a = 2$, and $b = 3$, a particular solution of the differential equation $\frac{d^2y}{dx^2} = 0$ is $y = 2x + 3$.

Solution of ODE:

➤ How to solve differential Equation $\frac{d^2 y}{d x^2} = 0$?

Solution:

$$\frac{d^2 y}{d x^2} = 0$$

$$\text{or, } \frac{d}{dx} \left(\frac{dy}{dx} \right) = 0$$

$$\text{or, } \int d \left(\frac{dy}{dx} \right) = a$$

$$\text{or, } \int du = a, \text{ where } u = \frac{dy}{dx}$$

$$\text{or, } u = a$$

$$\text{or, } \frac{dy}{dx} = a$$

$$\text{or, } \int dy = a \int dx + b \Rightarrow y = ax + b$$

which geometrically represents a straight line.

Or one may solve it in the following way :

$$\frac{d^2 y}{d x^2} = 0$$

$$\text{or, } \frac{d}{dx} \left(\frac{dy}{dx} \right) = 0$$

since the derivative of $\frac{dy}{dx}$ is zero;

so $\frac{dy}{dx} = \text{constant } a \text{ (say)}$

$$\text{or } \int dy = a \int dx + b \quad \text{or} \quad y = ax + b.$$

In fact $\frac{d^2 y}{d x^2} = 0$ is an ODE of order 2 and has as solution

Practice quiz: Classify differential equations

1. By checking all that apply, classify the following differential equation:

$$\frac{d^3y}{dx^3} + y \frac{d^2y}{dx^2} = 0$$

- a) first order
- b) second order
- c) third order
- d) ordinary
- e) partial
- f) linear
- g) nonlinear

2. By checking all that apply, classify the following differential equation:

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\psi}{d\xi} \right) = e^{-\psi}$$

- a) first order
- b) second order
- c) ordinary
- d) partial
- e) linear
- f) nonlinear

Practice quiz: Classify differential equations

3. By checking all that apply, classify the following differential equation:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = v \frac{\partial^2 u}{\partial x^2}$$

- a) first order
- b) second order
- c) ordinary
- d) partial
- e) linear
- f) nonlinear

4. By checking all that apply, classify the following differential equation:

$$a \frac{d^2 x}{dt^2} + b \frac{dx}{dt} + cx = 0$$

- a) first order
- b) second order
- c) ordinary
- d) partial
- e) linear
- f) nonlinear

Practice quiz: Classify differential equations

5. By checking all that apply, classify the following differential equation:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

- a) first order
- b) second order
- c) ordinary
- d) partial
- e) linear
- f) nonlinear

Solutions to the Practice quiz: Classify differential equations

1. c, d, g. Third order, ordinary, nonlinear.
2. b, c, f. Second order, ordinary, nonlinear.
3. b, d, f. Second order, partial, nonlinear.
4. b, c, e. Second order, ordinary, linear.
5. b, d, e. Second order, partial, linear.

Self Study

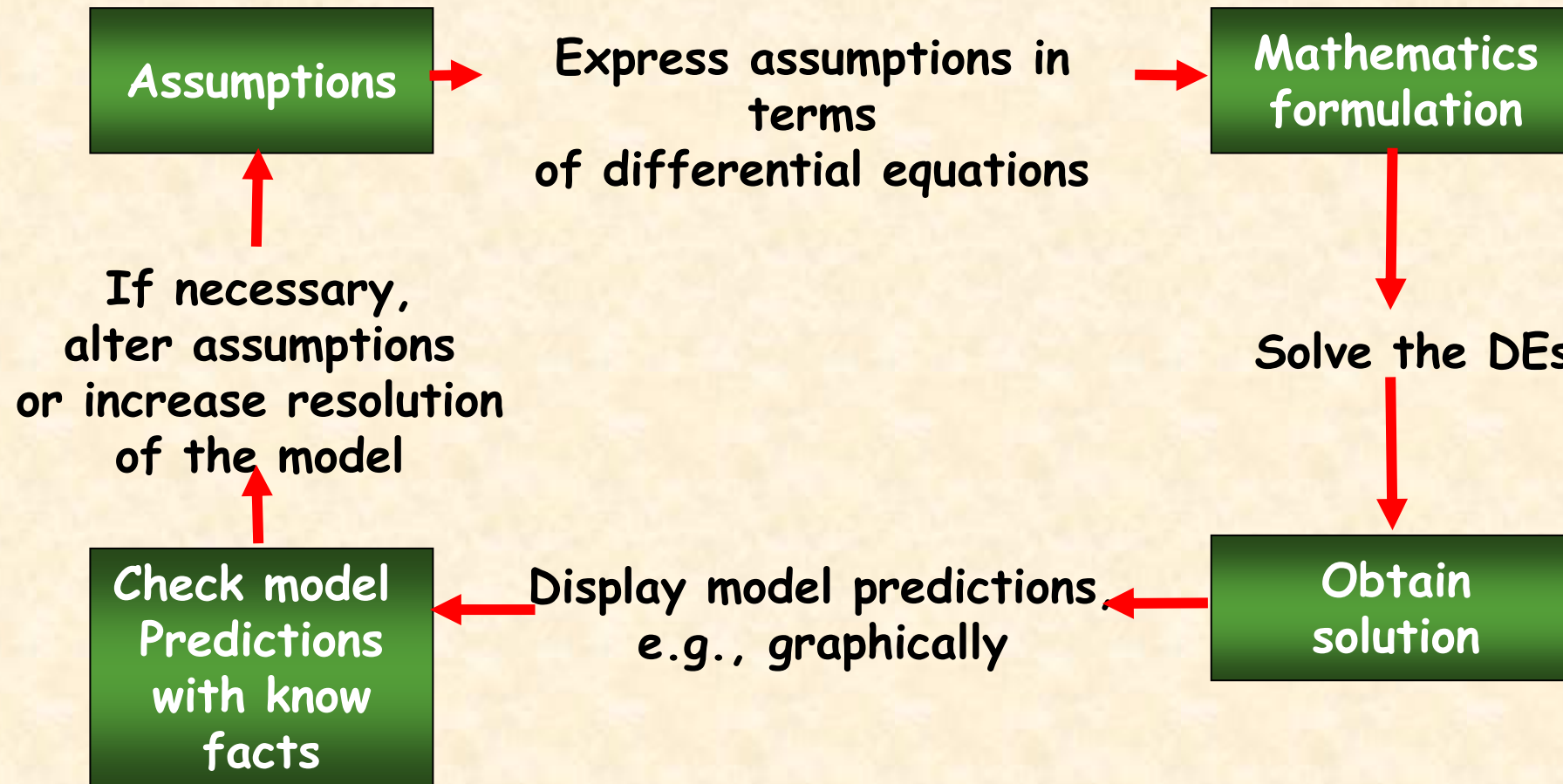
Classify the following differential equations are (i) Ordinary or Partial, (ii) linear or nonlinear and (iii) state order and degree each of the equation,

$$(i) 4 \frac{d^3 y}{dx^3} + \sin(y) \frac{d^2 y}{dx^2} + 5xy = 0$$

$$(ii) e^{y'} - xy'' + y = 0$$

$$(iii) \sqrt{h' + h} = \sin(\theta)$$

Formation of DE



Formation of ODE

An ordinary differential equation is formed by differentiating the equations and eliminating the arbitrary constants.

➤ Form an ODE corresponding to $y = ax + a^2$ also write down the order and degree of the obtained ODE.

Solution: $y = ax + a^2 \dots\dots\dots(i)$

Differentiating with respect to x , we get

$$\frac{d y}{d x} = a$$

Now putting the value of ' a ' into eq. (i), we get

$$y = x \frac{d y}{d x} + \left(\frac{d y}{d x} \right)^2$$

Which is the required ODE.

$\therefore$ Orde = 1, degree = 2.

Formation of ODE

- Form an ODE corresponding to $y = mx + c$ also write down the order and degree of the obtained ODE.

Solution: Given $y = mx + c$ (i)

Differentiating Eq. (i) with respect to x , we get

$$\frac{dy}{dx} = m \dots \dots \dots (ii)$$

Differentiating again Eq. (ii) with respect to x , we get

$$\frac{d^2 y}{dx^2} = 0$$

Which is the required ODE.

$\therefore$ Orde = 2, degree = 1.

Formation of ODE

➤ **Form an ODE corresponding to** $y = e^x (A \cos x + B \sin x)$.

Solution: Given $y = e^x (A \cos x + B \sin x) \dots \dots \dots (i)$

Differentiating Eq. (i) with respect to x , we get

$$\begin{aligned} \frac{d y}{d x} &= e^x (A \cos x + B \sin x) + e^x (-A \sin x + B \cos x) \\ &= y + e^x (-A \sin x + B \cos x) \dots \dots \dots (ii) \end{aligned}$$

Differentiating again Eq. (ii) with respect to x , we get

$$\begin{aligned} \frac{d^2 y}{d x^2} &= \frac{d y}{d x} + e^x (-A \sin x + B \cos x) + e^x (-A \cos x - B \sin x) \\ &= \frac{d y}{d x} + e^x (-A \sin x + B \cos x) - e^x (A \cos x + B \sin x) \\ &= \frac{d y}{d x} + \left(\frac{d y}{d x} - y \right) - y \quad \therefore \frac{d^2 y}{d x^2} - 2 \frac{d y}{d x} + 2 y = 0 \end{aligned}$$

Formation of ODE

1. Form an ODE corresponding to $y = a \cos x + b \sin x$ also write down the order and degree of the obtained ODE.

2. Derive an ODE corresponding to all circles lying in a plane.

Hints:

The equation of all circles lying in a plane is

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad \dots \dots \dots (i)$$

Which contains three arbitrary constant g , f and c .

3. Find the differential equation of the curves $xy = Ae^{2x} + Be^{-2x}$ for different values of A and B . State order and degree of the derived equation.

4. Show that $Ax^2 + By^2 = 1$ is the solution of $x \left[y \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx} \right)^2 \right] = y \frac{dy}{dx}$.

5. Find the differential equations of all circles passing through origin and having their centres on the X-axis.

Hints:

The equations of all circles passing through origin and having their centres on the X-axis is

$$x^2 + y^2 + 2gx = 0 \quad \dots \dots \dots (i)$$

Thanks a lot ...